PR 3 - Computer Vision Using OpenCV, YuNet and YOLOv8

All visual outputs from the notebook

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Figure 2: Task 1 - Colour vs grayscale image
Figure 3: Task 1 - Grayscale vs binary image
Figure 4: Task 1 - Erosion, dilation, opening and closing
Figure 5: Task 1 - Noisy binary image
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Figure 20: Task 5 - FPS comparison of detection techniques

Figure 1: Task 1 - Original sample image

Notebook cell 5

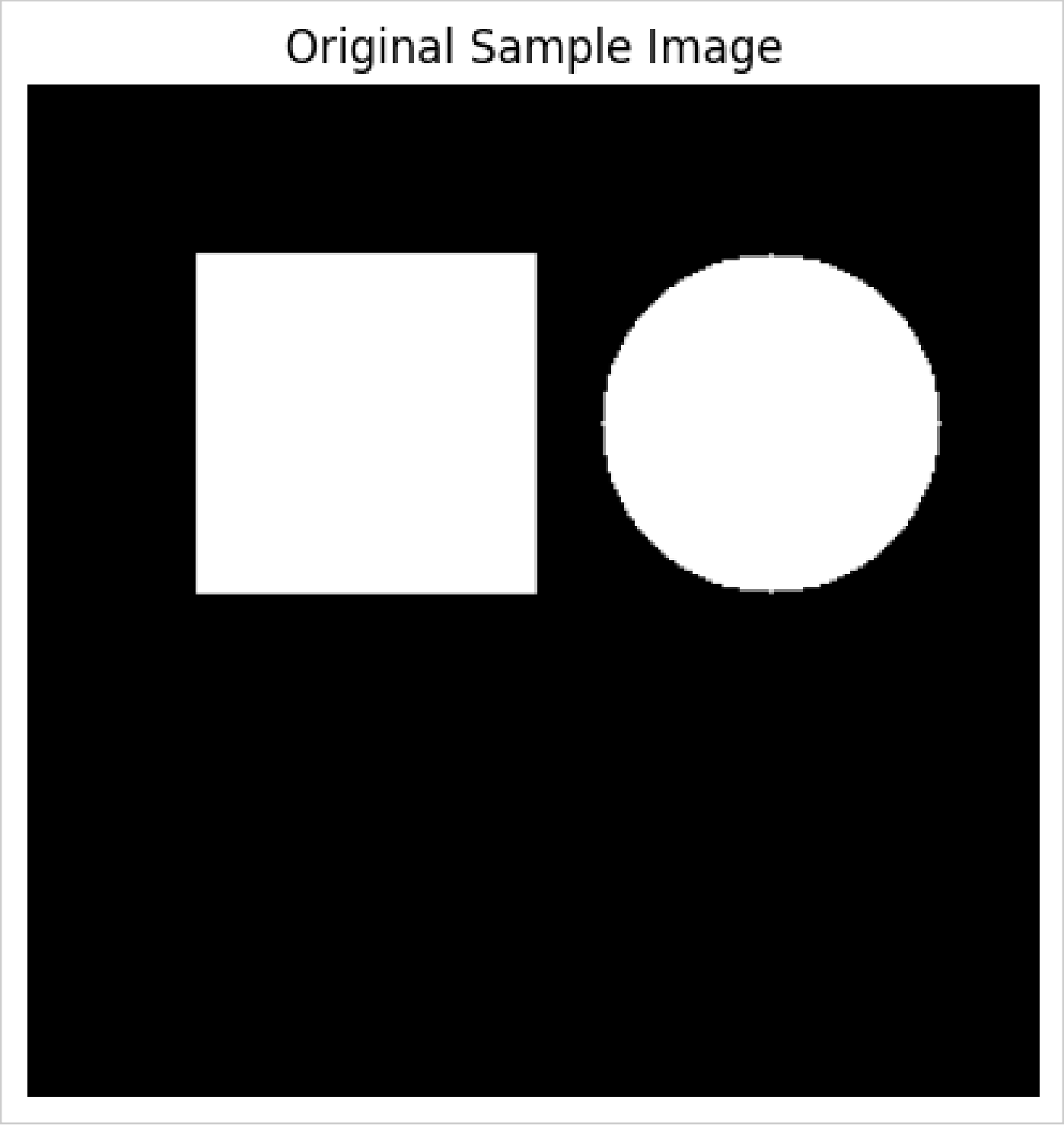


Figure 2: Task 1 - Colour vs grayscale image

Notebook cell 6

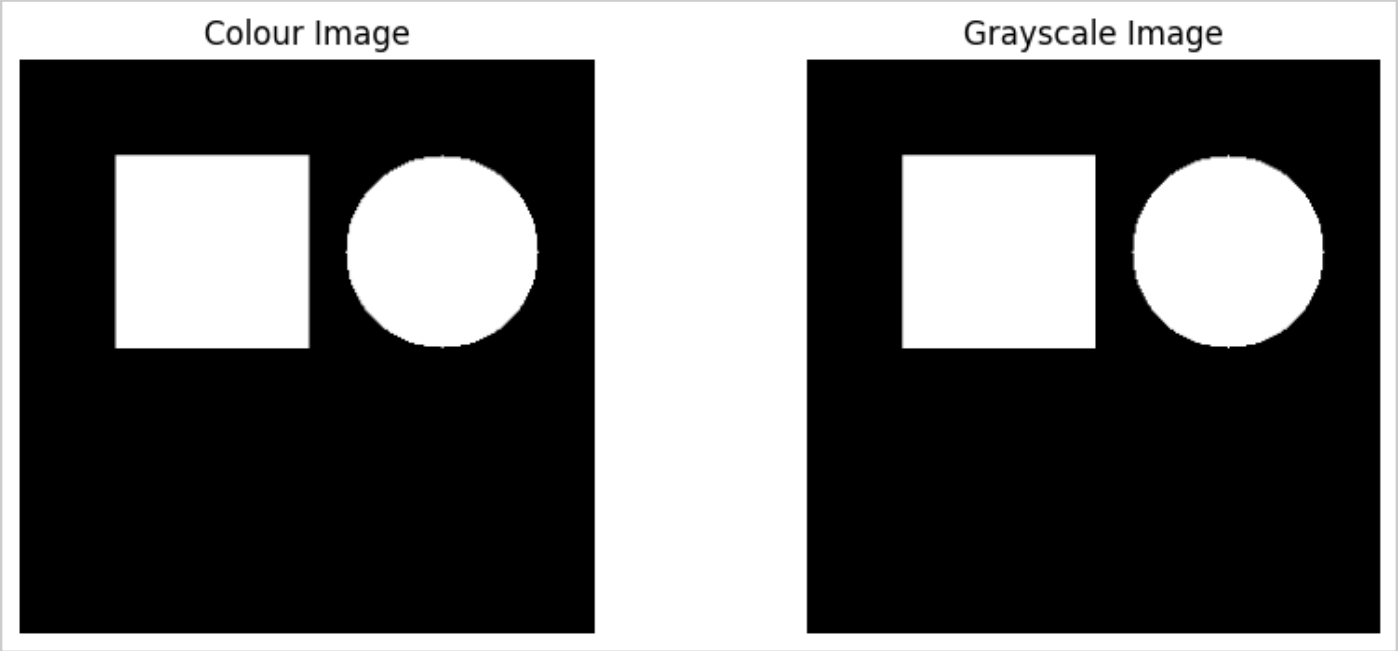


Figure 3: Task 1 - Grayscale vs binary image

Notebook cell 7

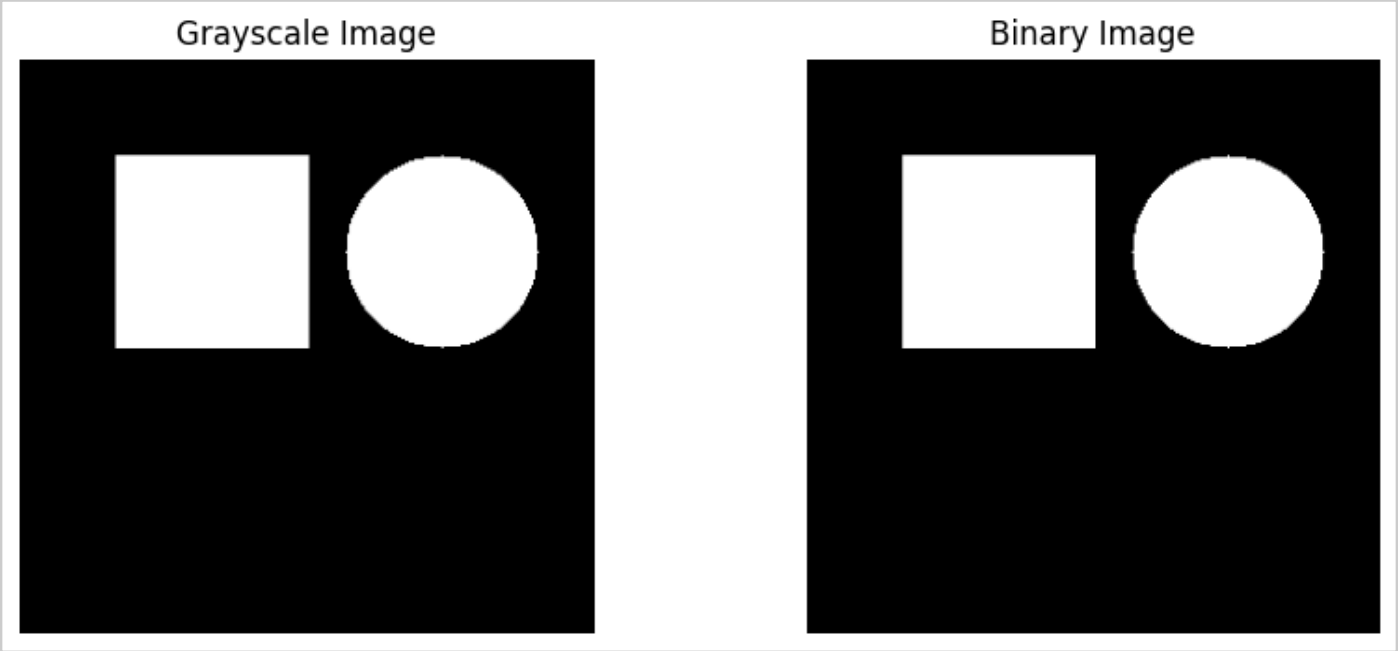


Figure 4: Task 1 - Erosion, dilation, opening and closing

Notebook cell 9

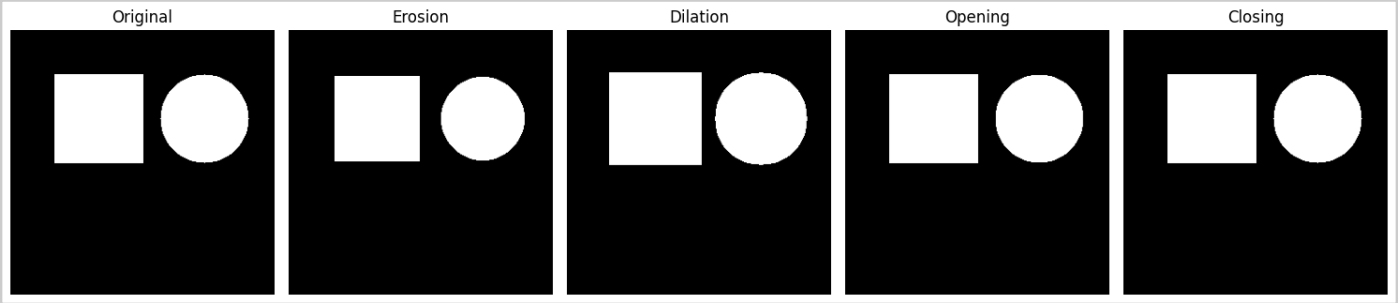


Figure 5: Task 1 - Noisy binary image

Notebook cell 10

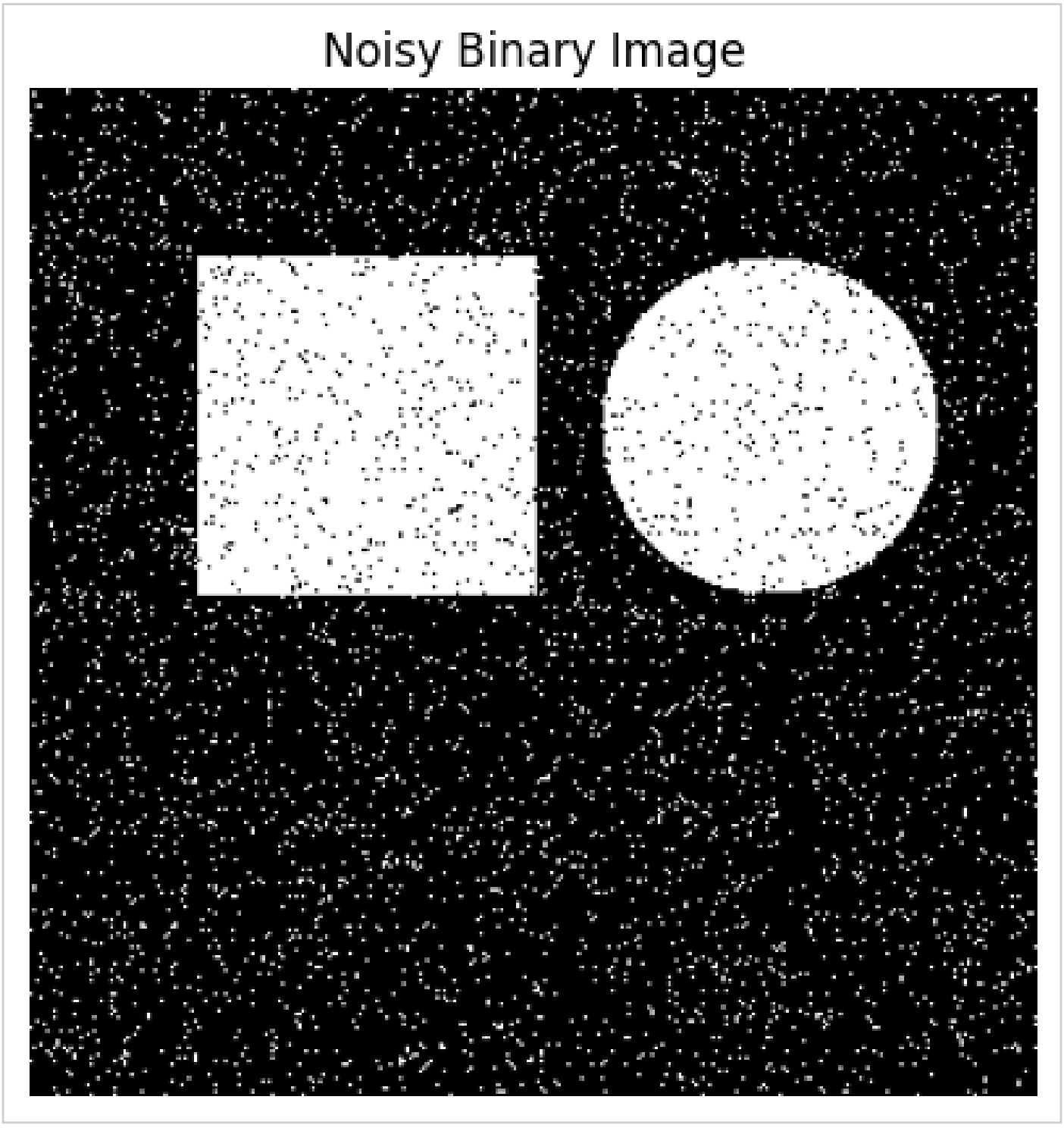


Figure 6: Task 1 - Opening with different kernel shapes and sizes

Notebook cell 11

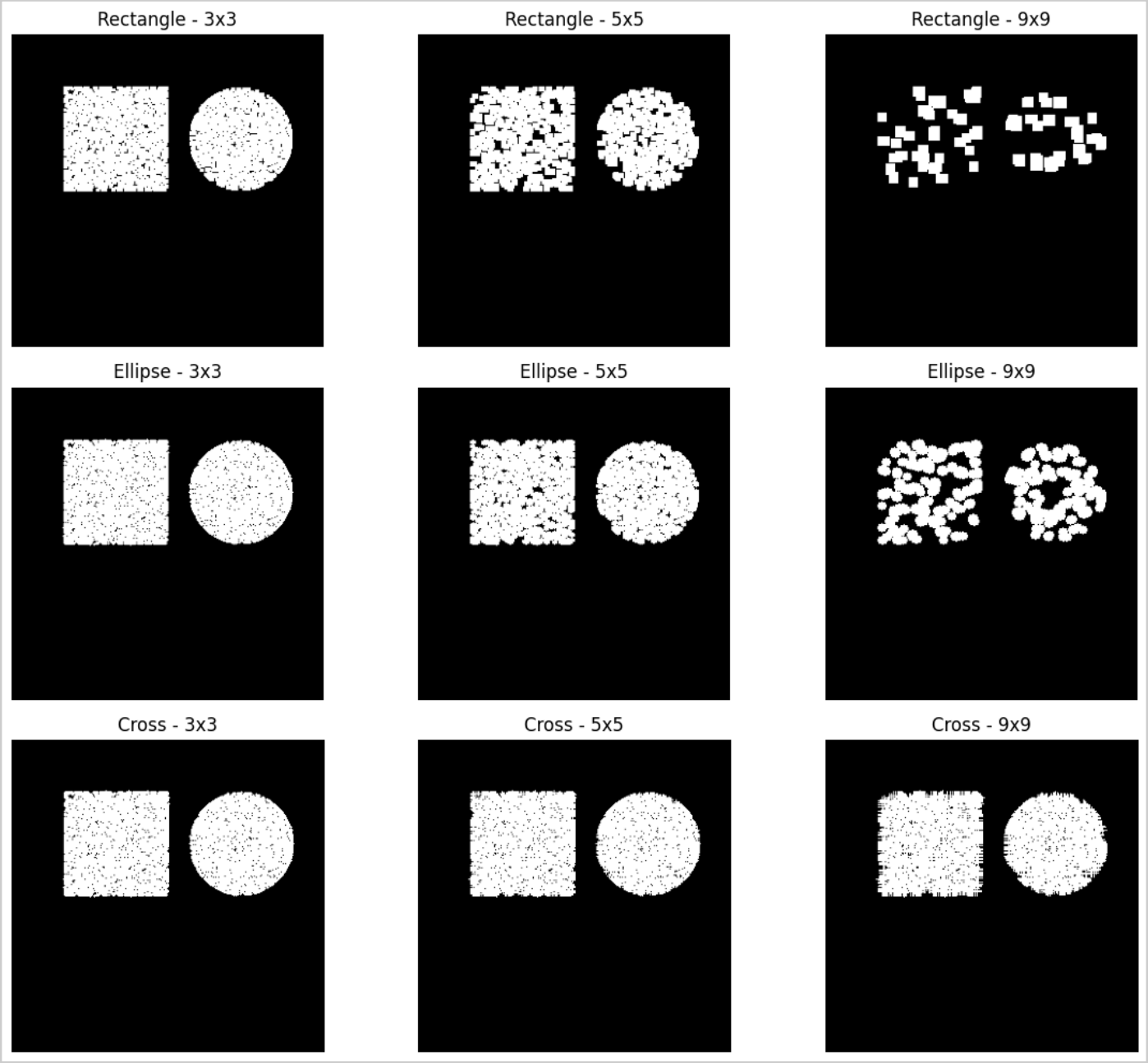


Figure 7: Task 1 - Closing with different kernel shapes and sizes

Notebook cell 12

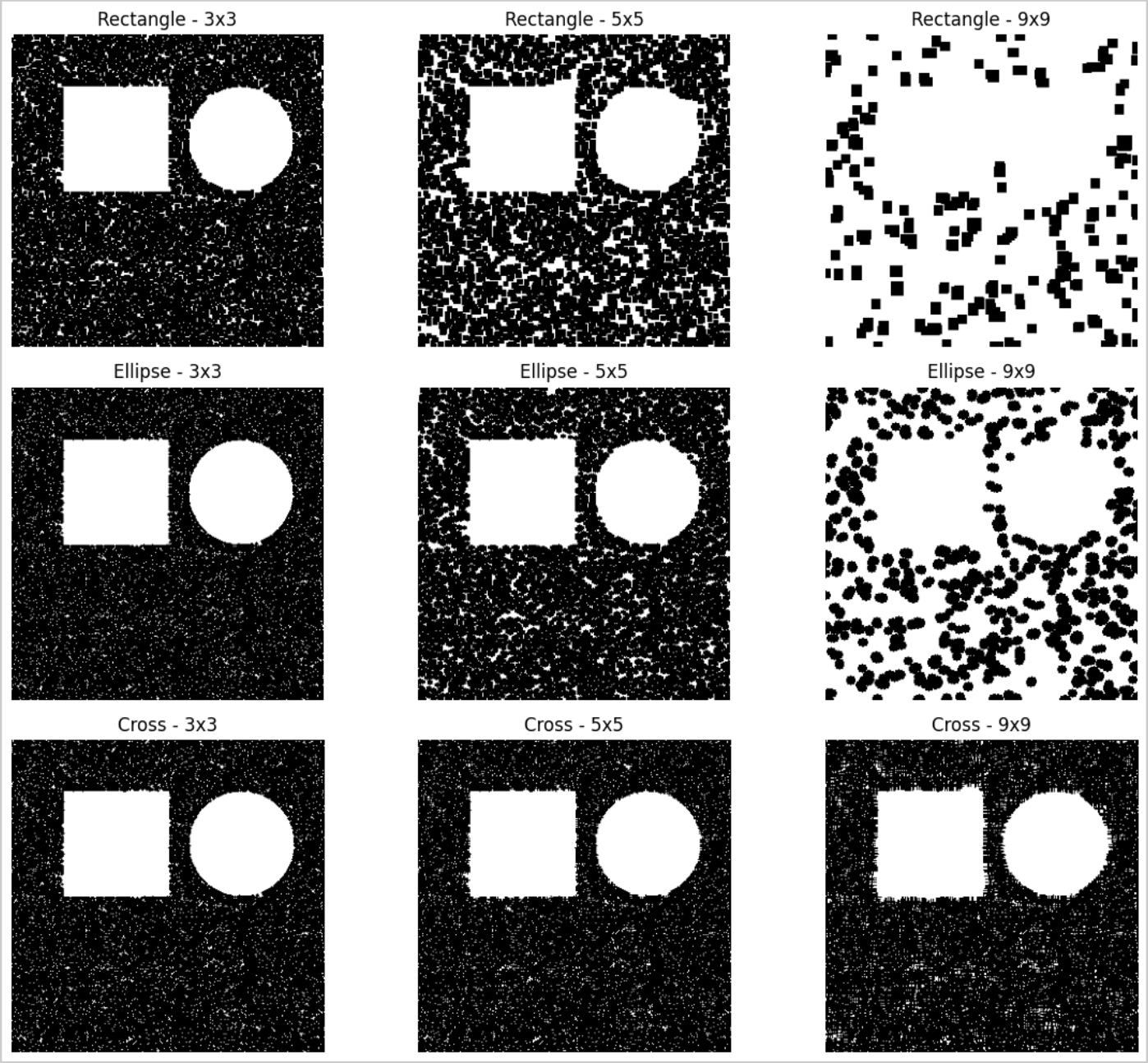


Figure 8: Task 1 - Saved morphology grid

Notebook cell 14

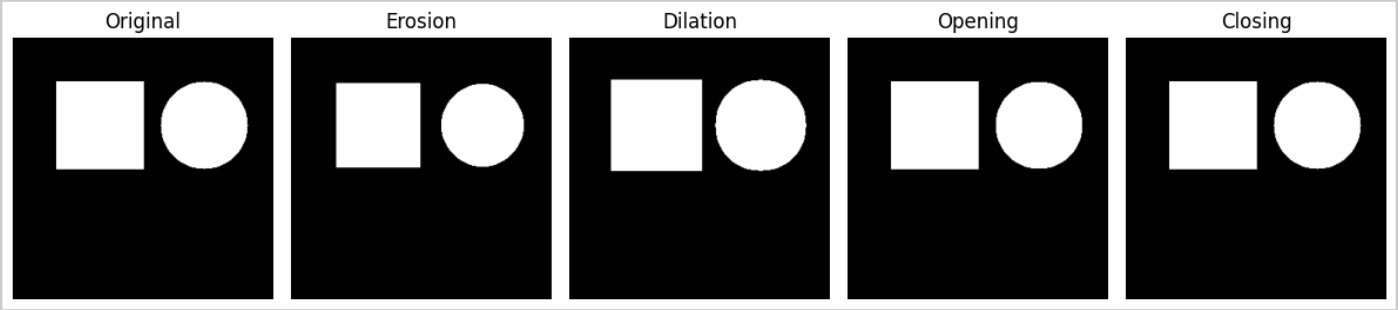


Figure 9: Task 2 - Circle and rectangle masks

Notebook cell 17

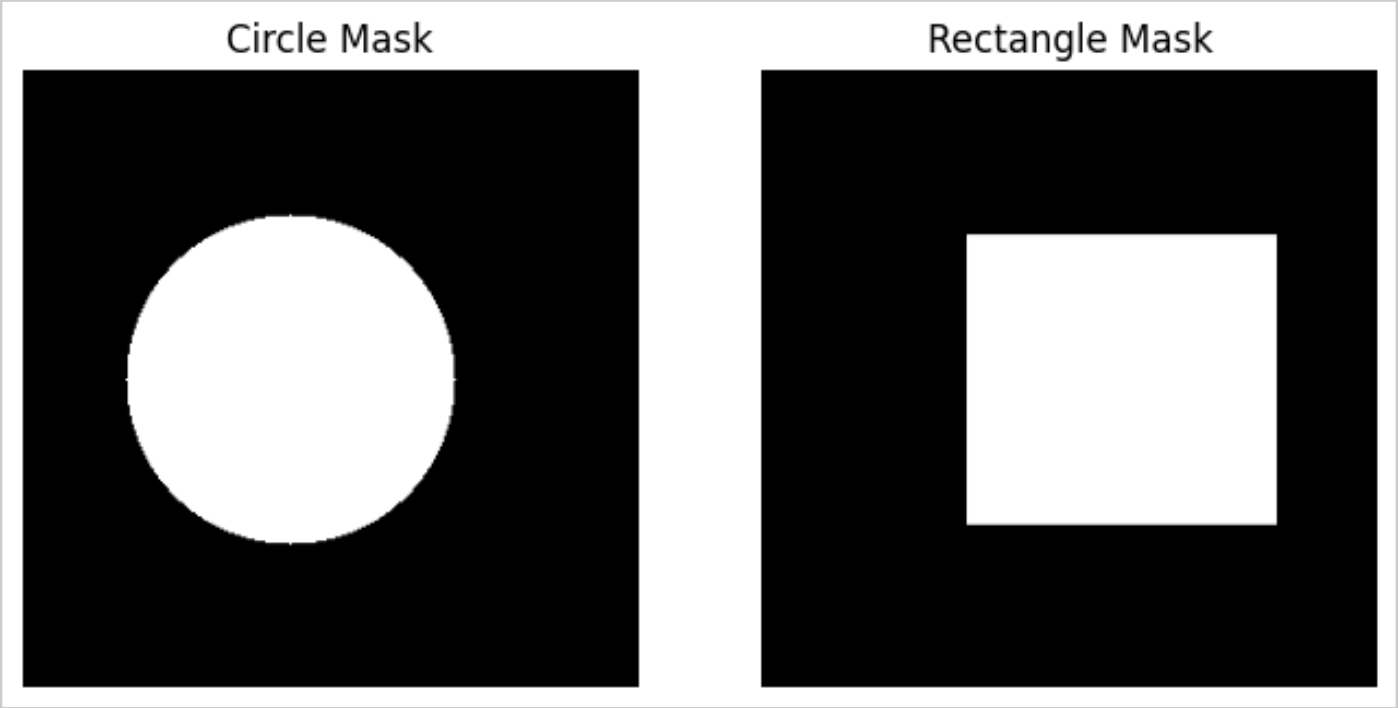


Figure 10: Task 2 - Bitwise AND, OR, XOR and NOT

Notebook cell 19

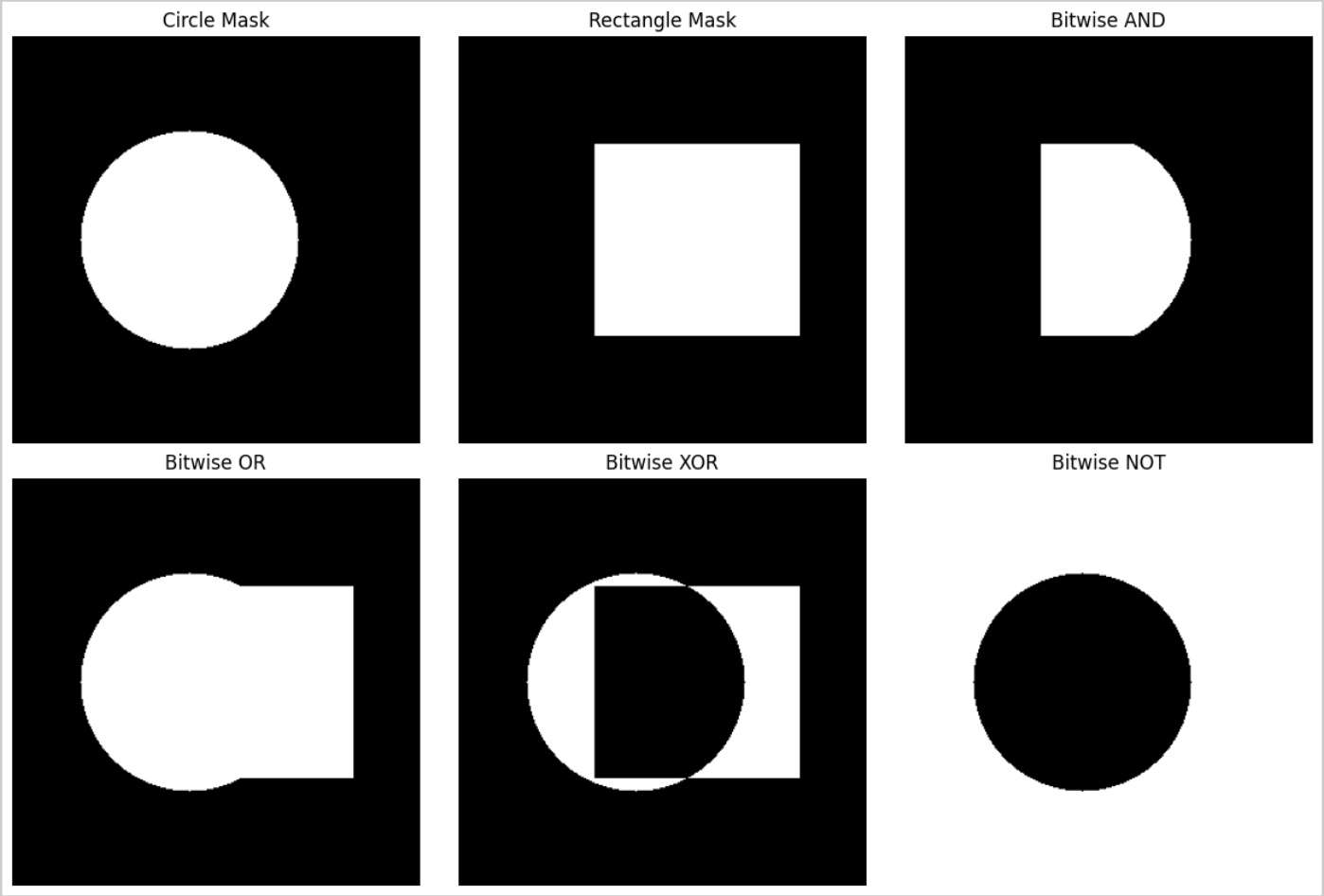


Figure 11: Task 2 - Original vs circular masked photo

Notebook cell 21

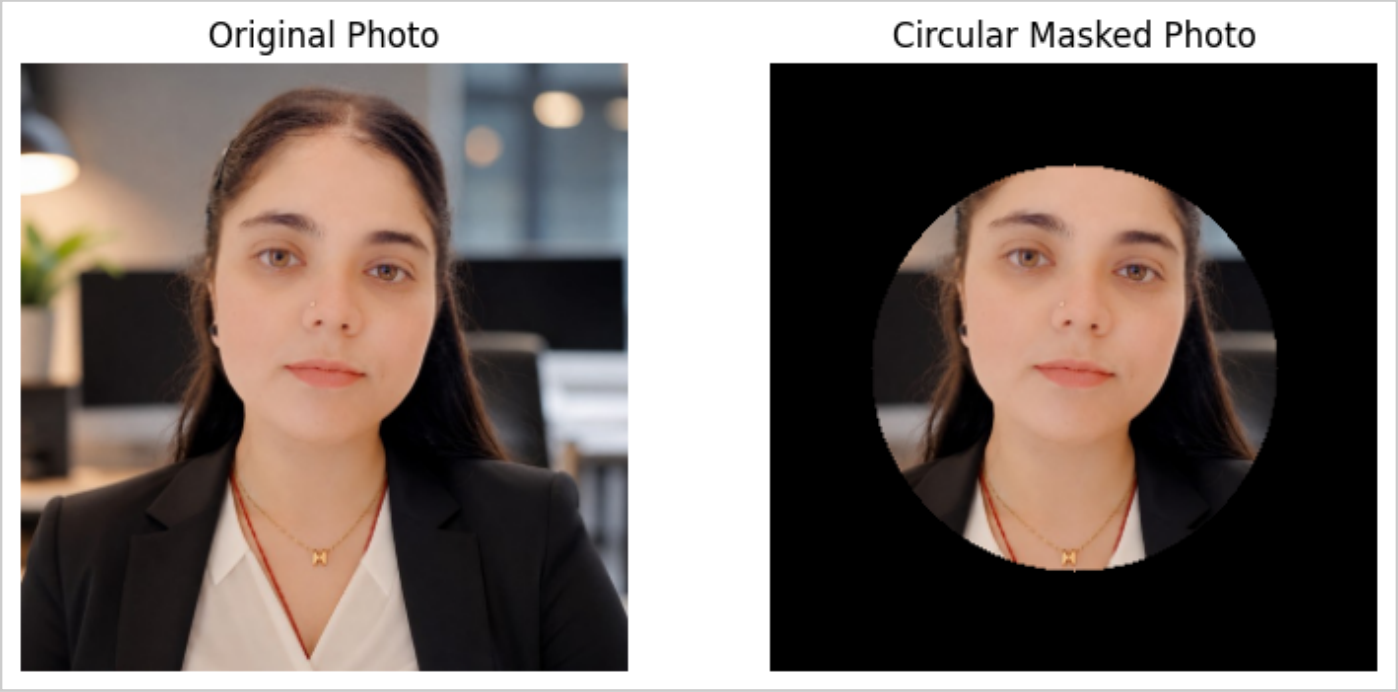


Figure 12: Task 2 - Grayscale histogram

Notebook cell 22

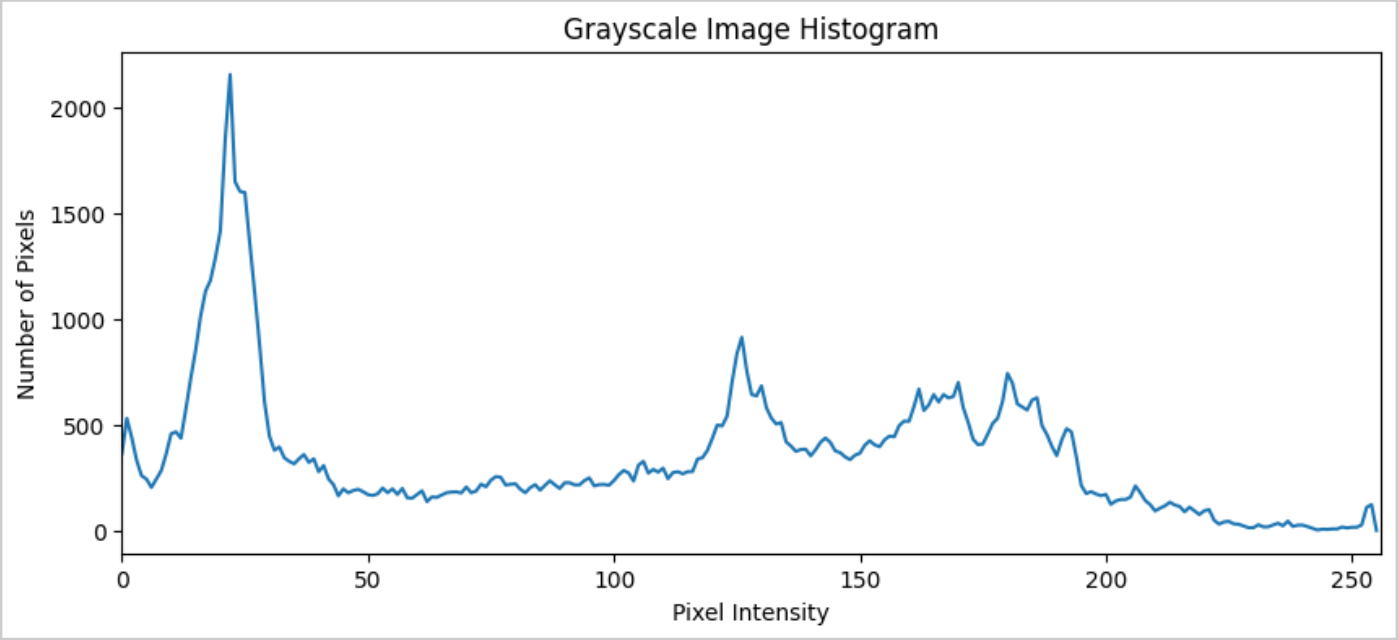


Figure 13: Task 2 - Colour (BGR) histogram

Notebook cell 23

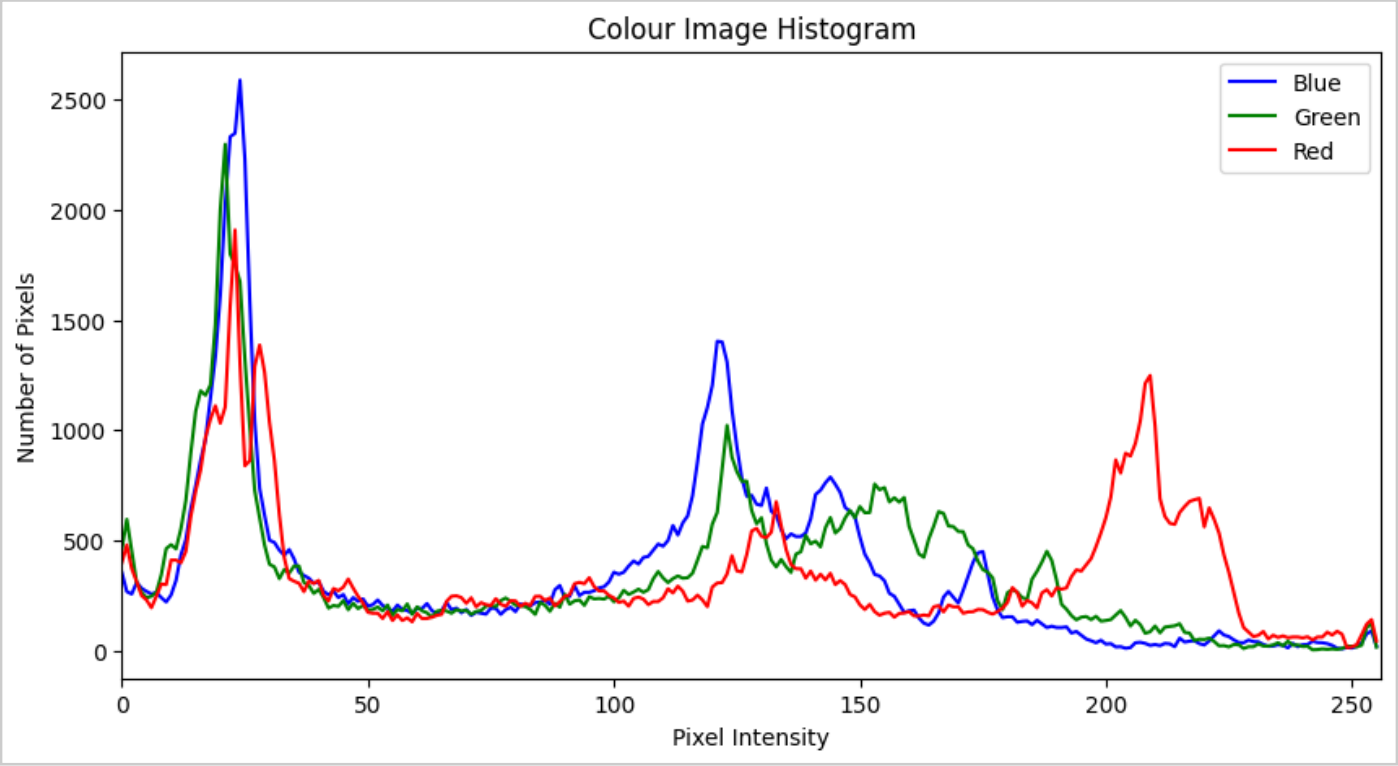


Figure 14: Task 2 - Brightness and contrast adjustment

Notebook cell 24

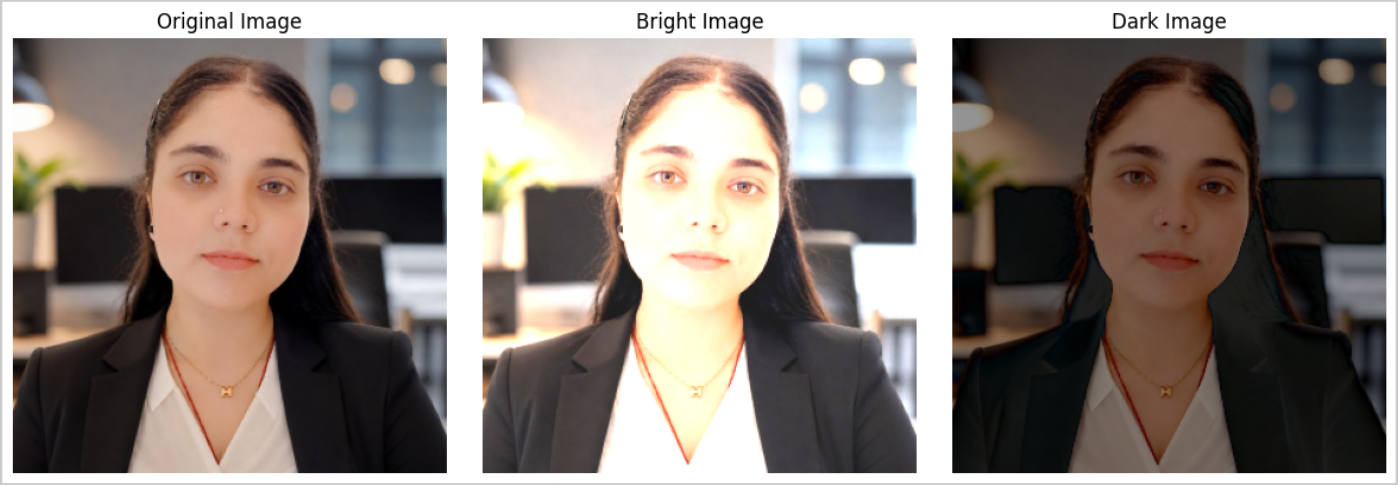


Figure 15: Task 2 - Histogram comparison (original, bright, dark)

Notebook cell 25

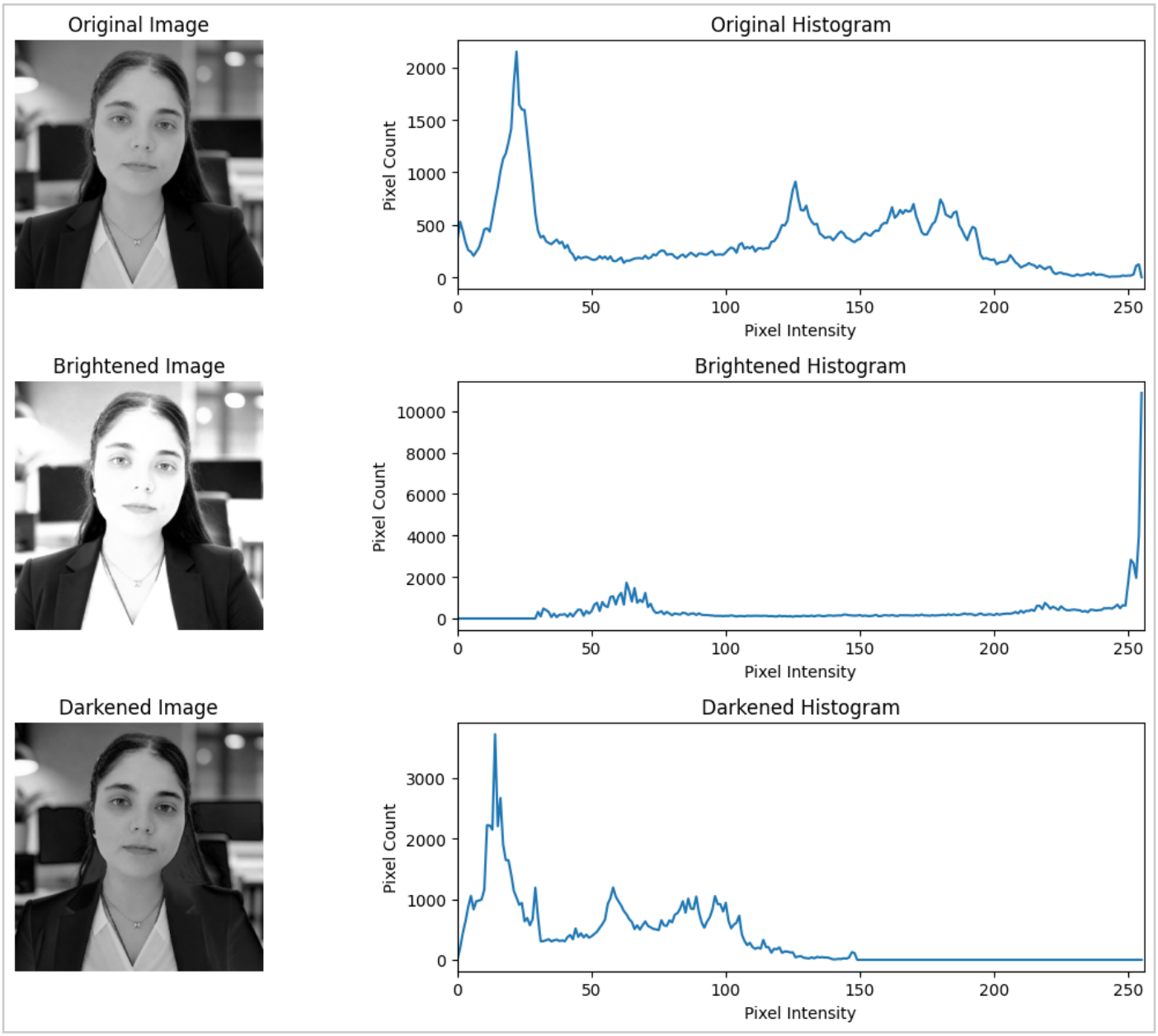


Figure 16: Task 3 - YuNet face detection result

Notebook cell 32

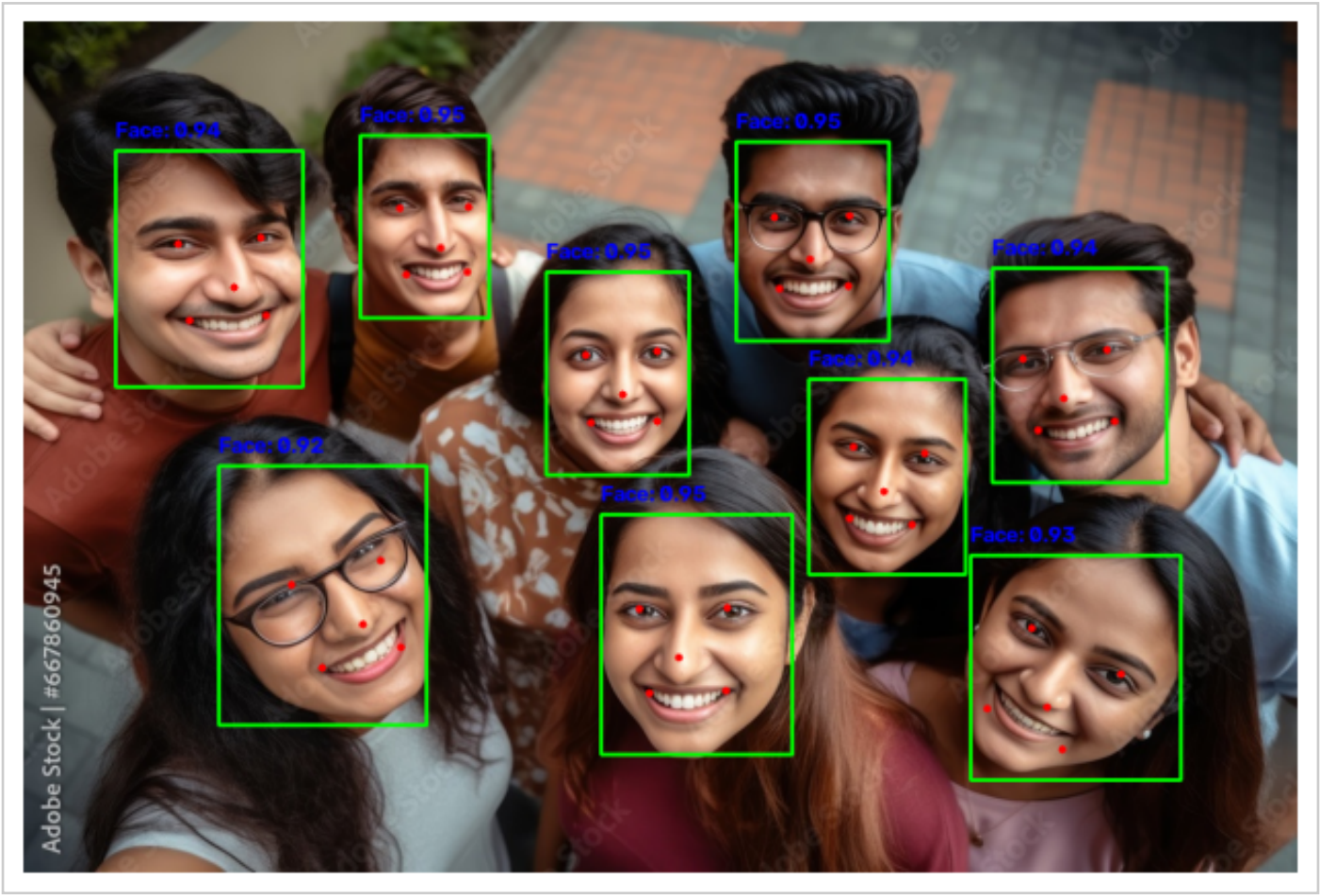


Figure 17: Task 3 - Gaussian blur on detected faces (privacy)

Notebook cell 34



Figure 18: Task 4 - YOLOv8 per-class detection summary

Notebook cell 47

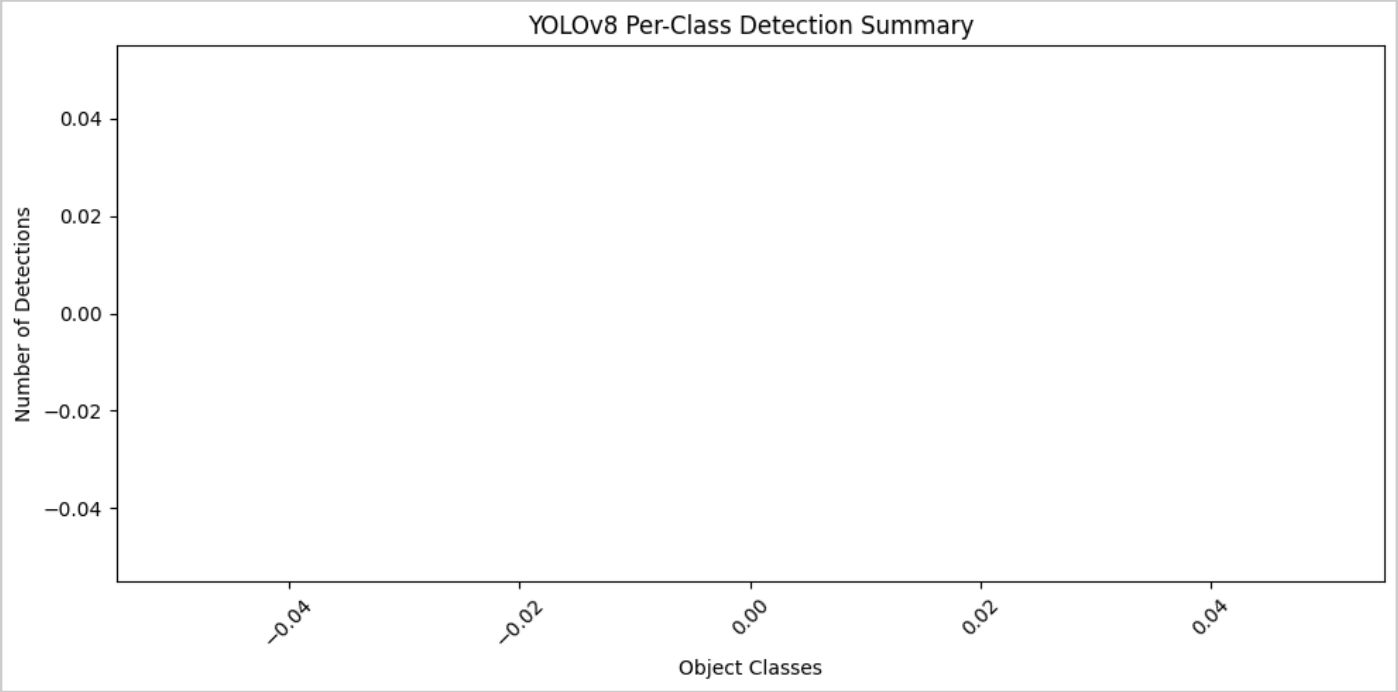


Figure 19: Task 4 - YOLOv8 per-class detection summary (re-run)

Notebook cell 50

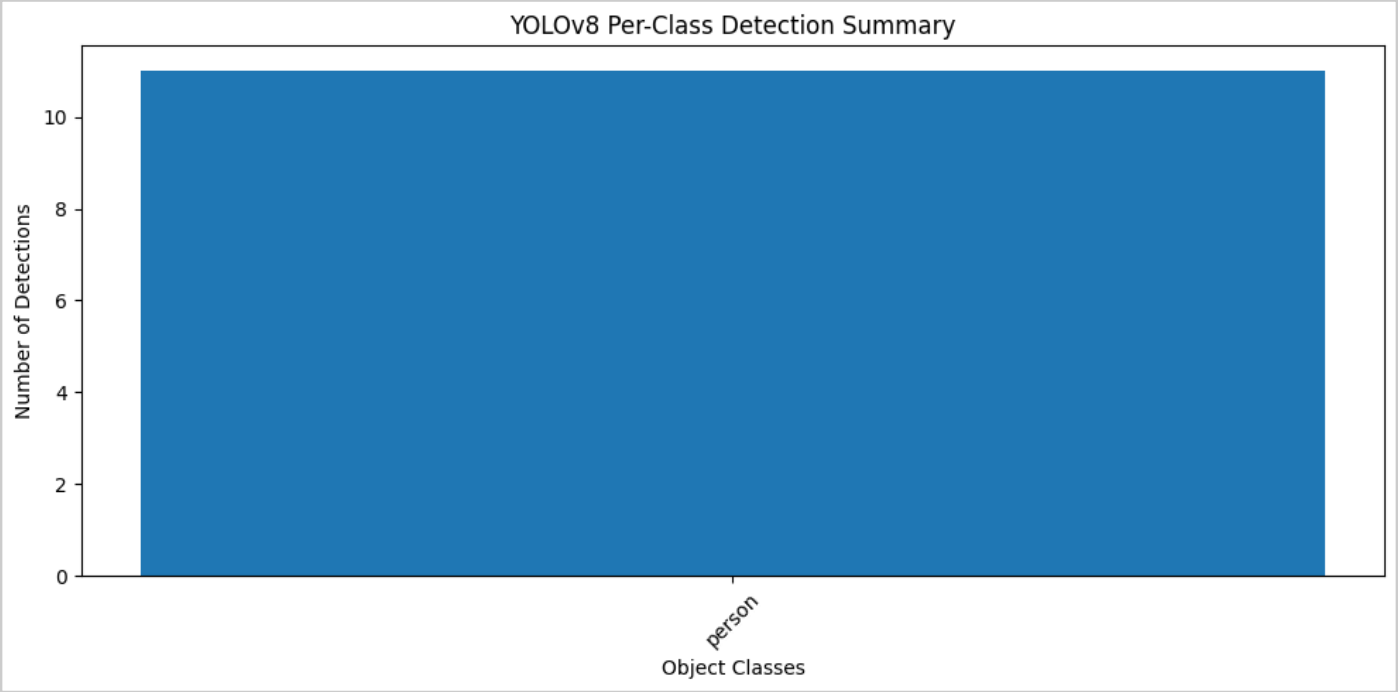


Figure 20: Task 5 - FPS comparison of detection techniques

Notebook cell 65

